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Linear and conic programming estimators in high dimensional errors-in-variables models

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Summary. We consider the linear regression model with observation error in the design. In this setting, we allow the number of covariates to be much larger than the sample size. Several new estimation methods have been recently introduced for this model. Indeed, the standard lasso estimator or Dantzig selector turns out to become unreliable when only noisy regressors are available, which is quite common in practice. In this work, we propose and analyse a new estimator for the errors-in-variables model. Under suitable sparsity assumptions, we show that this estimator attains the minimax efficiency bound. Importantly, this estimator can be written as a second-order cone programming minimization problem which can be solved numerically in polynomial time. Finally, we show that the procedure introduced by Rosenbaum and Tsybakov, which is almost optimal in a minimax sense, can be efficiently computed by a single linear programming problem despite non-convexities.

Keywords: Conic programming; Dantzig selector; Linear regression model; Matrix uncertainty selector; Prediction sensitivity

1. Introduction

We consider the regression model with observation error in the design:

$$\begin{aligned} y &= X\theta^* + \xi, \\ Z &= X + W. \end{aligned} \tag{1}$$

Here the random vector $y \in \mathbb{R}^n$ and the random $n \times p$ matrix Z are observed, the $n \times p$ matrix X is unknown, W is an $n \times p$ random-noise matrix, $\xi \in \mathbb{R}^n$ is a random-noise vector and $\theta^* \in \mathbb{R}^p$ is a vector of unknown parameters to be estimated. For example, the case where the entries of matrix X are missing at random can be reduced to this model. Such linear regressions with errors in both variables have been widely investigated in the literature; see for example Casella and Berger (1990), Cheng and van Ness (1999) and Kendall and Stuart (1973). Our work is

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different in that we consider the setting where the dimension p can be much larger than the sample size n , and θ^* is sparse.

It has been shown in Rosenbaum and Tsybakov (2010) that the presence of observation noise has severe consequences on the usual estimation procedures in the high dimensional setting. In particular, the lasso estimator and Dantzig selector turn out to be inaccurate and fail to identify the sparsity pattern of the vector θ^* . Rosenbaum and Tsybakov (2010) provided an alternative procedure, called the matrix uncertainty (MU) selector, which is robust to the presence of noise. The MU selector $\hat{\theta}^{\text{MU}}$ is defined as a solution of the minimization problem

$$\min \left\{ \|\theta\|_1 : \theta \in \Theta, \left\| \frac{1}{n} Z^T (y - Z\theta) \right\|_\infty \leq \mu \|\theta\|_1 + \tau \right\}, \quad (2)$$

where $\|\cdot\|_q$ denotes the l_q -norm for $1 \leq q \leq \infty$, Θ is a given convex subset of $\mathbb{R}^p$ characterizing the prior knowledge about θ^* and the constants $\mu \geq 0$ and $\tau \geq 0$ depend on the level of the noises W and ξ respectively. An extension of the MU selector to generalized linear models is discussed in Sørensen *et al.* (2014).

In Rosenbaum and Tsybakov (2013), a modification of the MU selector was suggested. It applies when W is a random matrix with independent and zero mean entries W_{ij} such that, for any $1 \leq j \leq p$, the sum of expectations

$$\sigma_j^2 = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(W_{ij}^2)$$

is finite and admits a data-driven estimator. This is for example the case in the model with missing data: $\tilde{Z}_{ij} = X_{ij}\eta_{ij}$, $i = 1, \dots, n$, $j = 1, \dots, p$, where, for each fixed $j = 1, \dots, p$, the factors η_{ij} , $i = 1, \dots, n$, are independent identically distributed Bernoulli random variables taking the value 1 with probability $1 - \pi_j$ and 0 with probability π_j , $0 < \pi_j < 1$. Indeed, this model can be rewritten under the form $Z_{ij} = X_{ij} + W_{ij}$, where $Z_{ij} = \tilde{Z}_{ij}/(1 - \pi_j)$ and $W_{ij} = X_{ij}\{\eta_{ij} - (1 - \pi_j)\}/(1 - \pi_j)$. Thus, in this model, the σ_j^2 satisfy

$$\sigma_j^2 = \frac{1}{n} \sum_{i=1}^n X_{ij}^2 \frac{\pi_j}{1 - \pi_j},$$

and it is easily shown that they admit good data-driven estimators $\hat{\sigma}_j^2$; see Rosenbaum and Tsybakov (2013). There are of course other examples where σ_j^2 can be accurately estimated from the data. One of them is related to repeated measurement models where the values of Z are available on a finer timescale. We refer here to examples given in Rosenbaum and Tsybakov (2010), such as portfolio replication. In the problem of portfolio replication, assuming for example that the entries of X are approximately constant on the daily scale, we can readily estimate the variances using the additional finer scale measurements of Z on the hourly scale.

The construction of this modified estimator is based on the following idea. We cannot use X in our estimation procedure since only its noisy version Z is available. In particular, the MU selector involves the matrix $Z^T Z/n$ instead of $X^T X/n$. Compared with $X^T X/n$, this matrix contains a bias that is induced by the diagonal entries of the matrix $W^T W/n$ whose expectations σ_j^2 do not vanish. Therefore, if the σ_j^2 can be estimated, a natural idea is to compensate this bias thanks to these estimates. This leads to a new estimator $\hat{\theta}^{\text{C}}$, called the compensated MU selector, and defined as a solution of the minimization problem

$$\min \left\{ \|\theta\|_1 : \theta \in \Theta, \left\| \frac{1}{n} Z^T (y - Z\theta) + \hat{D}\theta \right\|_\infty \leq \mu \|\theta\|_1 + \tau \right\}, \quad (3)$$

where $\hat{D}$ is the diagonal matrix with entries $\hat{\sigma}_j^2$, which are estimators of σ_j^2 , and $\mu \geq 0$ and $\tau \geq 0$ are constants chosen according to the level of the noise and the accuracy of the estimators $\hat{\sigma}_j^2$.

Several aspects of the compensated MU selector were studied in Rosenbaum and Tsybakov (2013), in particular the rates of convergence in l_q -norm, the prediction risk and the design of confidence intervals. One of the interests of this modification of the MU selector is that it enables us to obtain bounds for the estimation errors which are decreasing with the sample size n . This is in contrast with the case of the MU selector, where the bounds are small only if the noise W is small. For example, if θ^* is s sparse, it was shown in Rosenbaum and Tsybakov (2013) that, under appropriate assumptions,

$$|\hat{\theta}^C - \theta^*|_q \leq C s^{1/q} \left\{ \frac{\log(p)}{n} \right\}^{1/2} (|\theta^*|_1 + 1), \quad 1 \leq q \leq \infty, \quad (4)$$

with probability close to 1, where $C > 0$ is a constant independent of s , p , n and θ^* .

An alternative lasso-type estimator (non-convex programme) complemented by an iterative relaxation procedure was introduced in Loh and Wainwright (2012). This method requires knowledge of the exact value of $|\theta^*|_1$ (or of the property $|\theta^*|_1 \leq b\sqrt{s}$ for a constant b), and of an upper bound on $|\theta^*|_2$. Considering the setting where the entries of the regression matrix X are zero-mean sub-Gaussian, it was shown in Loh and Wainwright (2012) that if θ^* is s sparse, under appropriate assumptions, the resulting estimator $\hat{\theta}'$ satisfies

$$|\hat{\theta}' - \theta^*|_2 \leq C(\theta^*) s^{1/2} \left\{ \frac{\log(p)}{n} \right\}^{1/2} (|\theta^*|_2 + 1), \quad (5)$$

with probability close to 1, where $C(\theta^*) > 0$ depends on θ^* in a non-specified way. Related covariates selection results are reported in Sørensen *et al.* (2015). Chen and Caramanis (2012, 2013) proposed yet another method of estimation of θ^* , based on orthogonal matching pursuit. Their procedure needs the parameter s (the exact number of non-zero components of θ^*) as an input. Moreover, they imposed the additional assumption that the non-zero components θ_j^* of θ^* are sufficiently large:

$$|\theta_j^*| \geq c \left\{ \frac{\log(p)}{n} \right\}^{1/2} (|\theta^*|_2 + 1), \quad j = 1, \dots, p, \quad (6)$$

where $c > 0$ is a constant. Focusing as in Loh and Wainwright (2012) on the case where the entries of the regression matrix X are zero-mean sub-Gaussian, it was shown in Chen and Caramanis (2012, 2013) that the orthogonal matching pursuit estimator satisfies a bound analogous to condition (5) with constant $C(\theta^*) \equiv C > 0$ independent of θ^* , as well as a consistent support recovery property.

These recent developments shed light on errors-in-variables problems in high dimensional settings. However, they are not fully satisfying. Indeed, the following issues remain.

- From a practical viewpoint, the use of the above estimators can be intricate. In particular, the minimization problem (3) is not always a convex problem, and Rosenbaum and Tsybakov (2013) did not provide an algorithm enabling its solution in the general case. Although the methods that were suggested in Loh and Wainwright (2012) and Chen and Caramanis (2012, 2013) are computationally feasible, they need knowledge of the parameters $|\theta^*|_1$, $|\theta^*|_2$ or s , which are not available in practice.
- Although the bound (4) is more general than condition (5) (it holds for all q and for X that are not necessarily zero-mean sub-Gaussian), it is less accurate than bound (5) in

the case $q = 2$ assuming that bound (5) is established with $C(\theta^*) \equiv C > 0$ independent of θ^* . Indeed, $|\theta^*|_2$ is always smaller than $|\theta^*|_1$. For example, if all components of θ^* take the same value and θ^* is s sparse, then $|\theta^*|_2 = |\theta^*|_1 / \sqrt{s}$. In fact, the optimal rate of convergence in l_q -norm on the class of s -sparse vectors, as a function of s , p , n and the norms $|\theta^*|_r$, remains unknown. When $q = 2$ and X is zero-mean Gaussian, a minimax lower bound including the factor $|\theta^*|_2$ and not $|\theta^*|_1$ is stated without proof in Chen and Caramanis (2013). This, however, does not answer the question in the general situation.

The aim of this paper is to provide answers to the above two questions. The paper is organized as follows. After giving some definitions and assumptions in Sections 2 and 3, we introduce in Section 4 a new estimator $\hat{\theta}$ which is based on second-order cone programming and thus can be computed in polynomial time. We show that, under appropriate conditions, this estimator attains bounds of the form

$$|\hat{\theta} - \theta^*|_q \leq C s^{1/q} \left\{ \frac{\log(p)}{n} \right\}^{1/2} (|\theta^*|_2 + 1), \quad 1 \leq q \leq \infty, \quad (7)$$

with probability close to 1, where the constant C does not depend on s , p , n and θ^* . Contrary to the procedures of Chen and Caramanis (2012, 2013) and Loh and Wainwright (2012), this new estimator does not require knowledge of $|\theta^*|_1$, $|\theta^*|_2$ or s to be computed. We also do not need a lower bound condition such as condition (6) on the components of the target vector θ^* . Another difference from Chen and Caramanis (2012, 2013) and Loh and Wainwright (2012) is that our main results do not focus on zero-mean sub-Gaussian regression matrices X , but rather deal with deterministic matrices X commonly appearing in applications. As an easy consequence, we show that the results extend to random matrices X by using suitable deviation properties for the quantity

$$m_2 = \max_{j=1, \dots, p} \frac{1}{n} \sum_{i=1}^n X_{ij}^2,$$

where the X_{ij} are the entries of X , as well as checking a restricted-eigenvalue-type condition on the matrix $X^T X / n$. This extension is possible under the assumption that X is independent of ξ and W . Note that, under mild moment conditions, m_2 can be suitably controlled with high probability; see section 5 of the on-line supplementary material. Sufficient conditions for restricted-eigenvalue-type conditions are well known in the high dimensional setting; see Bickel *et al.* (2009).

Furthermore, in Section 5, we prove minimax lower bounds showing that no estimator can achieve faster rate than that given in expression (7), up to a logarithmic-in- s factor, uniformly on a class of s -sparse vectors. Finally, Section 6 provides simulation results that compare the conic estimator, the compensated MU selector and variants of the Dantzig selector. The results are in accordance with the theoretical findings.

Whereas the conic programming estimator solves a tractable convex minimization problem, the compensated MU selector is in general a non-convex programme. Section 1 in the on-line supplementary material is devoted to address this issue. We show that, under mild assumptions, the compensated MU selector can be reduced to convex programming. In fact, when $\Theta = \mathbb{R}^p$ or Θ is defined by linear constraints, it can even be written as a single linear programming problem, which is of course a computational advantage compared with the estimator based on conic programming. However, the rate of convergence of the compensated MU selector is suboptimal. The proofs are relegated to the appendices and some additional results and technical lemmas are given in the on-line supplementary material.

2. Assumptions on the model

In this section, we introduce the assumptions that will be used below to study the statistical properties of the estimators. Recall that, for $\gamma > 0$, the random variable η is said to be *sub-Gaussian with variance parameter γ^2* (or in brief γ *sub-Gaussian*) if, for all $t \in \mathbb{R}$,

$$\mathbb{E}\{\exp(t\eta)\} \leq \exp(\gamma^2 t^2 / 2).$$

A random vector $\zeta \in \mathbb{R}^p$ is said to be *sub-Gaussian with variance parameter γ^2* (or in brief γ *sub-Gaussian*) if the inner products (ζ, v) are γ sub-Gaussian for any $v \in \mathbb{R}^p$ with $|v|_2 = 1$. We shall consider the following assumptions.

Assumption 1. The matrix X is deterministic.

Assumption 2. The elements of the random vector ξ are independent zero-mean sub-Gaussian random variables with variance parameter σ^2 .

Assumption 3. The rows $w_i, i = 1, \dots, n$, of the noise matrix W are independent zero-mean sub-Gaussian random vectors with variance parameter σ_*^2 , and $\mathbb{E}(W_{ij}W_{ik}) = 0$ for all $1 \leq j < k \leq p$. Furthermore, W is independent of ξ .

Assumption 4. There are statistics $\hat{\sigma}_j^2$ such that, for any $\varepsilon > 0$, we have

$$\mathbb{P}\left\{\max_{j=1,\dots,p} |\hat{\sigma}_j^2 - \sigma_j^2| \geq b(\varepsilon)\right\} \leq \varepsilon, \quad (8)$$

where $b(\varepsilon) = c_b \{\log(c'_b p / \varepsilon) / n\}^{1/2}$ for some constants $c_b > 0$ and $c'_b > 0$.

Assumptions 1–3 are quite standard. Note that we do not assume independence of the components of each w_i . Examples of sufficient conditions for assumption 4 in the model with missing data are provided in Rosenbaum and Tsybakov (2013).

3. Sensitivities

It is well known (see for example Bickel *et al.* (2009)) that the performance of lasso- or Dantzig-selector-type estimators in high dimensional linear models is determined by specific characteristics of the Gram matrix

$$\Psi = \frac{1}{n} X^T X,$$

such as the restricted eigenvalue constants. We shall need similar characteristics here. Following Gautier and Tsybakov (2011), we define them in a more general form, so that the required property is a consequence of the restricted eigenvalue property whenever the latter is satisfied. For a vector θ in $\mathbb{R}^p$, we denote by θ_J the vector in $\mathbb{R}^p$ that has the same co-ordinates as θ on the set of indices $J \subset \{1, \dots, p\}$ and zero co-ordinates on its complement J^c . We denote by $|J|$ the cardinality of J .

For any $u > 0$ and any subset J of $\{1, \dots, p\}$, consider the cone

$$C_J(u) = \{\Delta \in \mathbb{R}^p : |\Delta_{J^c}|_1 \leq u |\Delta_J|_1\}.$$

The use of such cones to define the restricted eigenvalue constants and other related characteristics of the Gram matrix is standard in the literature on the lasso and Dantzig selector, starting from Bickel *et al.* (2009). For $q \in [1, \infty]$ and an integer $s \in [1, p]$, Gautier and Tsybakov (2011) defined the l_q -sensitivity as

$$\kappa_q(s, u) = \min_{J: |J| \leq s} \left(\min_{\Delta \in C_J(u): |\Delta|_q=1} |\Psi \Delta|_\infty \right).$$

In Gautier and Tsybakov (2011, 2013), it was shown that meaningful bounds for various types of errors in sparse linear regression can be obtained in terms of the sensitivities $\kappa_q(s, u)$. In particular, it is proved in Gautier and Tsybakov (2011) that the approach based on sensitivities is more general than that based on the restricted eigenvalue or the coherence condition. Indeed, under those assumptions,

$$\kappa_q(s, u) \geq cs^{-1/q},$$

for some constant $c > 0$, which implies the rate optimal bounds for the errors of the lasso and Dantzig selector estimators as in Bickel *et al.* (2009). For convenience, some properties of $\kappa_q(s, u)$ proved in Gautier and Tsybakov (2011) are summarized in section 3 of the on-line supplementary material.

In addition to $\kappa_q(s, u)$, we introduce a *prediction sensitivity* as follows:

$$\kappa_{\text{pr}}(s, u) = \min_{J: |J| \leq s} \left(\min_{\Delta \in C_J(u): |\Psi^{1/2} \Delta|_2=1} |\Psi \Delta|_\infty \right).$$

The sensitivity $\kappa_{\text{pr}}(s, u)$ is useful to establish convergence in the prediction norm with fast rates; see expression (17) in theorem 2 below. Such rates can be obtained under more general assumptions than rates of convergence in l_q -norm. A discussion of the case of repeated regressors is given in Belloni *et al.* (2014). The prediction sensitivity is closely related to the identifiability factor defined in Chernozhukov *et al.* (2013). Lemma 8 in section 3 of the on-line supplementary material shows that $\kappa_{\text{pr}}(s, u) > 0$ quite generally. Also, $\kappa_{\text{pr}}(s, u) \geq \sqrt{\kappa_1(s, u)}$; see lemma 7 in section 3 in the supplementary material.

4. Estimator based on conic programming

In this section, we introduce our conic-programming-based estimator $\hat{\theta}$. This estimator is computationally feasible and we provide upper bounds on its estimation and prediction errors. It will be shown in Section 5 that these bounds cannot be improved in a minimax sense. In what follows, we fix a (small) value $\varepsilon > 0$. The probabilities, with which the bounds on the estimation and prediction errors hold, will be of the form $1 - c\varepsilon$ for some $c > 0$.

To define the estimator $\hat{\theta}$, we consider the following minimization problem:

$$\begin{aligned} & \text{minimize } |\theta|_1 + \lambda t \text{ over } (\theta, t) \text{ such that} \\ & \theta \in \Theta, t \in \mathbb{R}^+, \left| \frac{1}{n} Z^T (y - Z\theta) + \hat{D}\theta \right|_\infty \leq \mu t + \tau, |\theta|_2 \leq t. \end{aligned} \quad (9)$$

Here λ , μ and τ are positive tuning constants, and Θ is a given subset of $\mathbb{R}^p$ characterizing the prior knowledge about θ . In the results below, we assume that

$$\lambda \in [\tfrac{1}{2}, 2].$$

The values τ and μ are of the order $\{\log(\rho/\varepsilon)/n\}^{1/2}$, up to factors depending only on m_2 and on the constants appearing in assumptions 1–4. More specifically, in the theory we take

$$\begin{aligned} \mu &= \delta'_1(\varepsilon) + \delta'_4(\varepsilon) + \delta_5(\varepsilon) + b(\varepsilon), \\ \tau &= \delta_2(\varepsilon) + \delta_3(\varepsilon) \end{aligned} \quad (10)$$

where $\delta_i(\varepsilon)$ and $\delta'_i(\varepsilon)$ are defined in lemmas 1 and 2 in Appendix A.

When $\Theta = \mathbb{R}^p$ or Θ is a subset of $\mathbb{R}^p$ defined by linear constraints, problem (9) is a second-order cone programming problem. Therefore it can be efficiently solved in polynomial time.

Let $(\hat{\theta}, \hat{t})$ be a solution of problem (9). We take $\hat{\theta}$ as estimator of θ^* . Under assumptions 1–4, it follows that the feasible set of the minimization problem (9) is not empty with high probability if ε is sufficiently small (see lemma 3 in Appendix B).

The following theorem, which is proved in Appendix B, is our main result about the statistical properties of the estimator $\hat{\theta}$ based on conic programming.

Theorem 1. Let assumptions 1–4 hold, and that the true parameter θ^* is s sparse and belongs to Θ . Let $\varepsilon > 0$, $1 \leq q \leq \infty$, and set μ and τ as in expression (10). Assume also that

$$\kappa_q(s, 1 + \lambda) \geq cs^{-1/q}, \quad (11)$$

for some constant $c > 0$ and that

$$s \leq c_1(\lambda^{-1} + \lambda)^{-1} \left\{ \frac{n}{\log(p/\varepsilon)} \right\}^{1/2}, \quad (12)$$

for some sufficiently small constant $c_1 > 0$. Then, with probability at least $1 - 8\varepsilon$,

$$|\hat{\theta} - \theta^*|_q \leq Cs^{1/q} \left\{ \frac{\log(c'p/\varepsilon)}{n} \right\}^{1/2} (|\theta^*|_2 + 1), \quad (13)$$

for some constants $C > 0$ and $c' > 0$ (here and in what follows we set $s^{1/\infty} = 1$).

Under the same assumptions, the prediction risk admits the following bound, with probability at least $1 - 8\varepsilon$:

$$\frac{1}{n} |X(\hat{\theta} - \theta^*)|_2^2 \leq Cs \frac{\log(c'p/\varepsilon)}{n} (|\theta^*|_2 + 1)^2. \quad (14)$$

The constants $C > 0$ and $c' > 0$ in inequalities (13) and (14) depend only on m_2 and on the constants appearing in assumptions 1–4.

Some remarks are in order here. Theorem 1 is established under the condition $\kappa_q(s, 1 + \lambda) \geq cs^{-1/q}$, which holds under standard assumptions on the matrix X . For example, it holds simultaneously for all q under the coherence assumption; see inequality (49) in the on-line supplementary material. For $1 \leq q \leq 2$ this condition follows from the restricted eigenvalue assumption; see inequalities (47) and (48) in the on-line supplementary material. It was shown in Rudelson and Zhou (2013) that the restricted eigenvalue assumption is satisfied with high probability for a large class of random matrices, including random matrices with zero-mean sub-Gaussian rows and non-trivial covariance structure, as well as matrices with zero-mean independent rows and uniformly bounded entries. This allows us to extend theorem 1 to random matrices X as follows. Assume that the conditional distribution of (ξ, W) given X is such that assumptions 2–4 are satisfied conditionally on X for some fixed constants c_b and c'_b and all $X \in \Omega$, where Ω is a given set of $n \times p$ matrices. For example, in the model with missing data that was considered in Section 1, this is so if $\Omega = \{X : \max_{j=1, \dots, p} (1/n) \sum_{i=1}^n X_{ij}^4 \leq m_4\}$ for some finite constant m_4 ; see Rosenbaum and Tsybakov (2013) and section 5 in the supplementary material. Fix positive constants ε , c , λ and m_2 and denote by $\mathcal{P}$ the class of probability distributions $\mathbf{P}_X$ on the set of $n \times p$ matrices X such that

$$\mathbf{P}_X \left\{ \kappa_q(s, 1 + \lambda) \geq cs^{-1/q}, \max_{j=1, \dots, p} \frac{1}{n} \sum_{i=1}^n X_{ij}^2 \leq m_2, X \in \Omega \right\} \geq 1 - \varepsilon. \quad (15)$$

Corollary 1. Let X be a random matrix with distribution $\mathbf{P}_X \in \mathcal{P}$, and let the above assumptions on the conditional distribution of (ξ, W) given X hold. Assume that the true parameter $\theta^* \in \Theta$ is s sparse and condition (12) holds and set μ and τ as in expression (10). Then, conditions (13) and (14) hold with probability at least $1 - 9\varepsilon$.

Although we do not pursue it here, theorem 1 implies results on the correct selection of the sparsity pattern via a thresholding procedure, in the same spirit as done in Lounici (2008).

Importantly, the bound (13) shows that the conic programming estimator is optimal in a minimax sense. Indeed, we give in Section 5 lower bounds for estimation errors which are in agreement with the upper bounds in condition (13). The conic programming estimator $\hat{\theta}$ achieves this rate with a computationally feasible procedure and does not need knowledge of the parameters $|\theta^*|_1$, $|\theta^*|_2$ or s .

Inspection of the proof reveals that, if condition (12) does not hold, the conclusions of theorem 1 remain valid provided that $|\theta^*|_2$ is replaced by $|\theta^*|_1$ in the bounds, thus leading to results that are analogous to those for the compensated MU selector. The next theorem formally states that. Note that the assumptions are different and somewhat weaker than in theorem 1.

Theorem 2. Let assumptions 1–4 hold. Let the true parameter θ^* be s sparse and belong to Θ . Let $\varepsilon > 0$, $1 \leq q \leq \infty$, and set μ and τ as in expression (10). Then, with probability at least $1 - 8\varepsilon$,

$$|\hat{\theta} - \theta^*|_q \leq \frac{C}{\kappa_q(s, 1 + \lambda)} \left\{ \frac{\log(c' p / \varepsilon)}{n} \right\}^{1/2} (|\theta^*|_1 + 1), \quad (16)$$

for some constants $C > 0$ and $c' > 0$. Under the same assumptions, the prediction risk admits the following bound, with probability at least $1 - 8\varepsilon$:

$$\frac{1}{n} |X(\hat{\theta} - \theta^*)|_2^2 \leq \frac{C}{\kappa_{\text{pr}}^2(s, 1 + \lambda)} \frac{\log(c' p / \varepsilon)}{n} (|\theta^*|_1 + 1)^2. \quad (17)$$

Furthermore, under no assumption on X , with the same probability, we have the following ‘slow rate’ bound:

$$\frac{1}{n} |X(\hat{\theta} - \theta^*)|_2^2 \leq C \left\{ \frac{\log(c' p / \varepsilon)}{n} \right\}^{1/2} (|\theta^*|_1^2 + |\theta^*|_1). \quad (18)$$

The constants $C > 0$ and $c' > 0$ in inequalities (16)–(18) depend only on m_2 and on the constants appearing in assumptions 1–4.

There are three different results in theorem 2. The bound (16) is based on the l_q -sensitivity measures without the sparsity condition (12) and recovers the rates of the compensated MU selector. The second result (17) presents a prediction rate but the prediction sensitivity allows for more general designs. Finally, the last result in theorem 2 provides a slow rate of convergence that requires no assumption on the design matrix.

5. Minimax lower bounds for arbitrary estimators

In this section, we show that the rates of convergence that are obtained in theorem 1 are optimal (up to a logarithmic-in- s term) in a minimax sense for all estimators over the intersection of the class of s -sparse vectors and the l_2 -sphere:

$$B_0(s) = \{\theta : |\theta|_0 \leq s\}$$

and

$$S_2(R) = \{\theta : |\theta|_2 = R\},$$

where $R > 0$. Defining the parameter set as the intersection $\Theta = B_0(s) \cap S_2(R)$ is motivated by the presence of both s and $|\theta^*|_2$ in the upper bounds of theorem 1. Note that considering a deterministic X means that X is a nuisance parameter of the model. Thus, in the definition of the minimax risk, one should take the maximum not only over Θ but also over a class of possible matrices X . More generally, one can deal with random X and with the maximum over a class of distributions of X . We shall follow this approach with the class of distributions $\mathcal{P}$ introduced in Section 4. The result of corollary 1 corresponding to bound (13) can be written as

$$\sup_{\mathbf{P}_X \in \mathcal{P}} \sup_{\theta \in B_0(s) \cap S_2(R)} \mathbb{P}_{X, \theta} \left[|\hat{\theta} - \theta|_q \geq C s^{1/q} \left\{ \frac{\log(c' p / \varepsilon)}{n} \right\}^{1/2} (R+1) \right] \leq 9\varepsilon, \quad (19)$$

where, for $\theta \in \mathbb{R}^p$, we denote by $\mathbb{P}_{X, \theta}$ the probability measure of the pair (y, Z) satisfying model (1). Our aim now is to prove the reverse inequality to inequality (19) valid for all estimators. For this, instead of the maximum over all $\mathbf{P}_X \in \mathcal{P}$, it suffices in principle to consider one particular distribution $\mathbf{P}_X$. We choose it to be the distribution of Gaussian matrix X with independent identically distributed rows and positive definite covariance matrix. This enables us, in addition, to obtain minimax optimality when restricting the class $\mathcal{P}$ to one such Gaussian distribution only. With high probability, these matrices satisfy the restricted eigenvalue condition, which implies the inequality $\kappa_q(s, 1 + \lambda) \geq c s^{-1/q}$ under the probability in inequality (15) (see for example Rudelson and Zhou (2013) for details). Also, we shall assume that ξ and W are Gaussian with independent identically distributed entries. In summary, we make the following assumption.

Assumption 5. The elements of the triplet (ξ, X, W) are jointly independent. The components of ξ and W are independent identically distributed Gaussian zero-mean random variables with positive variances σ^2 and σ_*^2 respectively. The rows of X are independent identically distributed Gaussian zero-mean random vectors with covariance matrix $\Sigma > 0$.

Denote by $\lambda_{\min}^\Sigma$ and $\lambda_{\max}^\Sigma$ the smallest and largest eigenvalues of Σ . The next theorem, which is proved in Appendix C, provides the desired minimax lower bound.

Theorem 3. Let $p \geq 2$, $1 \leq q \leq \infty$, $2 \leq s \leq p$, and $R > 0$. Let assumption 5 hold and $s \log(p/s)/n \leq \bar{c} R^2 / (R^2 + 1)$ for some constant $\bar{c} > 0$. Then there are constants $c > 0$ and $c' > 0$, depending only on q , σ^2 , σ_*^2 , $\bar{c}$, $\lambda_{\min}^\Sigma$ and $\lambda_{\max}^\Sigma$, such that

$$\inf_{\hat{T}} \sup_{\theta \in B_0(s) \cap S_2(R)} \mathbb{P}_{X, \theta} \left[|\hat{T} - \theta|_q \geq c s^{1/q} \left\{ \frac{\log(p/s)}{n} \right\}^{1/2} (R+1) \right] > c', \quad (20)$$

where $\inf_{\hat{T}}$ denotes the infimum over all estimators, and we set $s^{1/\infty} = 1$.

6. Monte Carlo study

In this section, we briefly illustrate the empirical performance of the estimators that were discussed above. We consider the proposed conic programming estimator with $\lambda = 0.5, 0.75, 1$ (denoted as Conic(λ)) and the compensated MU selector CompMU. To have benchmarks, we also compute the (unfeasible) Dantzig selector which knows X , Dantzig X , and the Dantzig selector that uses only Z , Dantzig Z , ignoring the errors-in-variables issue.

The simulation study uses the following data-generating process:

$$y_i = x_i^T \theta^* + \xi_i, \\ z_i = x_i + w_i.$$

Here, ξ_i, w_i and x_i are independent and $\xi_i \sim \mathcal{N}(0, \sigma^2)$, $w_i \sim \mathcal{N}(0, \sigma_*^2 I_{p \times p})$ and $x_i \sim \mathcal{N}(0, \Sigma)$ where $I_{p \times p}$ is the identity matrix and Σ is a $p \times p$ matrix with elements $\Sigma_{ij} = \rho^{|i-j|}$. We set $\sigma = 0.128$, $\sigma_* = 0.45$ and $\rho = 0.25$. For simplicity, we assume that σ_* and σ are known and we set $\hat{D} = D = \sigma_*^2 I_{p \times p}$. The penalty parameters are set as $\tau = \mu = \sigma \{\log(p/\varepsilon)/n\}^{1/2}$ for $\varepsilon = 0.05$. We consider two choices for the vector of unknown parameters θ^* . The first choice is $\theta^* = (1, 1, 1, 1, 0, \dots, 0)^T$, which captures the case where the coefficients are well separated from 0. The second choice is $\theta^* = (1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, 0, \dots, 0)^T$, which represents the situation where θ^* is sparse with components that are not necessarily well separated from 0. The results are based on 100 replications. The implementation of these estimators was based on interior point methods which might not be suitable for large instances. The average running times of the six estimators were within a factor of 2. For $p = 10, 50$ the average running time of all six estimators was below 1 s, for $p = 100$ all estimators were below 3 s and for $p = 500$ the average running times were between 40 and 100 s. We note that Conic(1) and CompMU had very comparable running times, both Dantzig X and Dantzig Z had faster running times than the other estimators, and Conic(0.5) and Conic(0.75) require slightly more time to be computed.

Table 1 reports the simulation results in the case $\theta^* = (1, 1, 1, 1, 0, \dots, 0)^T$. As expected, the performance of all the estimators deteriorates as p grows but only slightly. Also, the (unfeasible) estimator based on the Dantzig selector that observes X outperforms all feasible options. The estimator that ignores the errors-in-variables issue appears with a higher bias leading to the worse result in terms of root-mean-square error and prediction risk. The performance of the feasible estimators that are discussed in this paper is between these two benchmarks. The three conic estimators exhibit a better result than the compensated MU selector when $p = 10, 50$.

Table 1. Simulation results for the first choice of θ^* †

Method	Results for first θ^* , $n = 300$ and $p = 10$			Results for first θ^* , $n = 300$ and $p = 50$		
	Bias	RMSE	PR	Bias	RMSE	PR
Conic(0.5)	0.0838151	0.1846383	0.1710643	0.0955776	0.2245046	0.2170111
Conic(0.75)	0.0838151	0.1846383	0.1710643	0.0953689	0.2250219	0.2176691
Conic(1)	0.0838151	0.1846383	0.1710643	0.0956858	0.2253614	0.2180705
CompMU	0.1566904	0.2191588	0.2225818	0.1840462	0.2362394	0.2507162
Dantzig X	0.0265486	0.0321528	0.0349530	0.0301636	0.0349420	0.0386731
Dantzig Z	0.2952845	0.3300527	0.3645317	0.3078976	0.4166192	0.4174840
Results for first θ^* , $n = 300$ and $p = 100$			Results for first θ^* , $n = 300$ and $p = 500$			
Conic(0.5)	0.1101178	0.2556778	0.2474407	0.1668239	0.2656095	0.2635298
Conic(0.75)	0.0943678	0.2711839	0.2606997	0.1425789	0.2846916	0.2789745
Conic(1)	0.0942906	0.2734750	0.2631424	0.1276741	0.3121221	0.3093194
CompMU	0.1910509	0.2539411	0.2658907	0.2052520	0.2657204	0.2772154
Dantzig X	0.0317776	0.0366155	0.0403419	0.0352309	0.0403134	0.0448000
Dantzig Z	0.3081669	0.4994041	0.4652972	0.3536668	0.6865989	0.5921541

†For each estimator we provide the average bias (denoted as Bias), average root-mean-squared error RMSE and average prediction risk denoted as PR.

Table 2. Simulation results for the second choice of θ^* †

Method	Results for second θ^* , $n = 300$ and $p = 10$			Results for second θ^* , $n = 300$ and $p = 50$		
	Bias	RMSE	PR	Bias	RMSE	PR
Conic(0.5)	0.0564816	0.1020763	0.0987642	0.0684162	0.1236275	0.1223037
Conic(0.75)	0.0564816	0.1020763	0.0987642	0.0682720	0.1229000	0.1219398
Conic(1)	0.0564816	0.1020763	0.0987642	0.0683291	0.1227749	0.1218898
CompMU	0.0839431	0.1171633	0.1204765	0.1007774	0.1318303	0.1396494
Dantzig X	0.0265486	0.0321528	0.0349530	0.0301636	0.0349420	0.0386731
Dantzig Z	0.1885828	0.2024266	0.2138763	0.1949159	0.2314964	0.2319208
	Results for second θ^* , $n = 300$ and $p = 100$			Results for second θ^* , $n = 300$ and $p = 500$		
	Bias	RMSE	PR	Bias	RMSE	PR
Conic(0.5)	0.0714621	0.1374637	0.1349551	0.0945558	0.1472914	0.1477198
Conic(0.75)	0.0713670	0.1378301	0.1353203	0.0824510	0.1589884	0.1565416
Conic(1)	0.0716242	0.1381810	0.1357000	0.0783823	0.1682841	0.1658849
CompMU	0.1063728	0.1405472	0.1479579	0.1131005	0.1477336	0.1545960
Dantzig X	0.0317776	0.0366155	0.0403419	0.0352309	0.0403134	0.0448000
Dantzig Z	0.1978958	0.2536633	0.2432222	0.2152972	0.3145349	0.2766815

†For each estimator we provide the average bias (denoted as Bias), average root-mean-squared error RMSE and average prediction risk denoted as PR.

For the larger dimensions $p = 100, 500$, their performance becomes similar to that of the compensated MU selector. Nonetheless, the conic estimator with $\lambda = 0.5$ is slightly better than all the other feasible estimators. We also note that, for the small dimension $p = 10$, all three conic estimators give the same results. The reason is that the conic constraint was not active in the simulations for $p = 10$ so the estimators were the same for the range of λ under consideration. This was not so for $p = 50, 100, 500$. These findings are very much aligned with the theoretical properties of each estimator and confirm that the effect of errors in variables can be substantial.

Table 2 reports the results for the second choice of θ^* , where the coefficients are not well separated from 0. They are qualitatively the same as before. The results confirm the robustness of the conclusions with respect to possible model selection errors which are unavoidable when the coefficients are not well separated from zero.

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Appendix A: Bounds on stochastic error terms

In this appendix, we give upper bounds on the stochastic error terms appearing in the main results. In what follows, D is a diagonal matrix with diagonal elements σ_j^2 , $j = 1, \dots, p$, and, for a square matrix A ,

we denote by $\text{Diag}\{A\}$ the matrix with the same dimensions as A , the same diagonal elements as A and all off-diagonal elements equal to 0. The following lemma is proved in Rosenbaum and Tsybakov (2013).

Lemma 1. Let $0 < \varepsilon < 1$ and let assumptions 1–3 hold. Then, with probability at least $1 - \varepsilon$,

$$\begin{aligned} \left| \frac{1}{n} X^T W \right|_{\infty} &\leq \delta_1(\varepsilon), & \left| \frac{1}{n} X^T \xi \right|_{\infty} &\leq \delta_2(\varepsilon), & \left| \frac{1}{n} W^T \xi \right|_{\infty} &\leq \delta_3(\varepsilon), \\ \left| \frac{1}{n} (W^T W - \text{Diag}\{W^T W\}) \right|_{\infty} &\leq \delta_4(\varepsilon), & \left| \frac{1}{n} \text{Diag}\{W^T W\} - D \right|_{\infty} &\leq \delta_5(\varepsilon), \end{aligned}$$

where $\delta_1(\varepsilon) = \tilde{\delta}(\varepsilon, \sigma_*, 2p^2)$, $\delta_2(\varepsilon) = \tilde{\delta}(\varepsilon, \sigma, 2p)$, $\delta_3(\varepsilon) = \delta_5(\varepsilon) = \bar{\delta}(\varepsilon, 2p)$ and $\delta_4(\varepsilon) = \bar{\delta}(\varepsilon, p(p-1))$ and, for an integer N ,

$$\begin{aligned} \tilde{\delta}(\varepsilon, a, N) &= a \left\{ \frac{2m_2 \log(N/\varepsilon)}{n} \right\}^{1/2}, \\ \bar{\delta}(\varepsilon, N) &= \max \left[\gamma_0 \left\{ \frac{2 \log(N/\varepsilon)}{n} \right\}^{1/2}, \frac{2 \log(N/\varepsilon)}{t_0 n} \right], \end{aligned}$$

with γ_0 and t_0 positive constants depending only on σ and σ_* .

We now give the second lemma.

Lemma 2. Let $0 < \varepsilon < 1$, $\theta^* \in \mathbb{R}^p$ and let assumptions 1–3 hold. Then, with probability at least $1 - \varepsilon$,

$$\left| \frac{1}{n} X^T W \theta^* \right|_{\infty} \leq \delta'_1(\varepsilon) |\theta^*|_2, \quad (21)$$

where $\delta'_1(\varepsilon) = \sigma_* \{2m_2 \log(2p/\varepsilon)/n\}^{1/2}$. In addition, with probability at least $1 - \varepsilon$,

$$\left| \frac{1}{n} (W^T W - \text{Diag}\{W^T W\}) \theta^* \right|_{\infty} \leq \delta'_4(\varepsilon) |\theta^*|_2, \quad (22)$$

where

$$\delta'_4(\varepsilon) = \max \left[\gamma_2 \left\{ \frac{2 \log(2p/\varepsilon)}{n} \right\}^{1/2}, \frac{2 \log(2p/\varepsilon)}{t_2 n} \right],$$

with γ_2 and t_2 positive constants depending only on σ_* .

Proof. If $\theta^* = 0$, the result is obvious. So we assume that $\theta^* \neq 0$. Let $v = \theta^* / |\theta^*|_2$. We can write

$$\left| \frac{1}{n} X^T W \theta^* \right|_{\infty} = |\theta^*|_2 \max_{j=1, \dots, p} \left| \frac{1}{n} \sum_{i=1}^n X_{ij}(w_i, v) \right|, \quad (23)$$

where $(w_i, v) = \sum_{k=1}^p W_{ik} v_k$ and we denote by W_{ik} and v_k the elements of the matrix W and the vector v respectively. By assumption 3, the random variable (w_i, v) is sub-Gaussian with variance parameter σ_*^2 . Using this together with the independence of the w_i for different i , we obtain that, for all $t \in \mathbb{R}$,

$$\mathbb{E} \left[\exp \left\{ \frac{t}{n} \sum_{i=1}^n X_{ij}(w_i, v) \right\} \right] = \prod_{i=1}^n \mathbb{E} \left[\exp \left\{ \frac{t}{n} X_{ij}(w_i, v) \right\} \right] \leq \prod_{i=1}^n \exp \left(\frac{\sigma_*^2 t^2 X_{ij}^2}{2n^2} \right) \leq \exp \left(\frac{\sigma_*^2 m_2 t^2}{2n} \right).$$

Thus, the random variable

$$\eta_j = \frac{1}{n} \sum_{i=1}^n X_{ij}(w_i, v)$$

is γ_1 sub-Gaussian with $\gamma_1 = \sigma_* (m_2/n)^{1/2}$. This implies the classical tail bound

$$\mathbb{P}(|\eta_j| \geq \delta) \leq 2 \exp \left\{ -\delta^2 / (2\gamma_1^2) \right\},$$

for any $\delta > 0$. This, together with equation (23) and the union bound, yields inequality (21).

To prove inequality (22), we shall use the following fact (see for example lemma 5.14 in Vershynin (2012)): if η is a sub-Gaussian random variable with variance parameter γ^2 , then η^2 is subexponential, i.e. if there are constants $\gamma_0 = \gamma_0(\gamma)$ and $t_0 = t_0(\gamma)$ such that

$$\mathbb{E}\{\exp(t\eta^2)\} \leq \exp(\gamma_0^2 t^2/2), \quad |t| \leq t_0. \quad (24)$$

Analogously to equation (23), we obtain

$$\left| \frac{1}{n} (W^T W - \text{Diag}\{W^T W\}) \theta^* \right|_\infty = |\theta^*|_2 \max_{j=1, \dots, p} |\eta'_j|, \quad (25)$$

where

$$\eta'_j = \frac{1}{n} \sum_{i=1}^n W_{ij} \sum_{k=1, k \neq j}^p W_{ik} v_k.$$

Now, for all $t \in \mathbb{R}$, we have

$$\mathbb{E}\{\exp(t\eta'_j)\} = \prod_{i=1}^n \mathbb{E}\left\{\exp\left(\frac{tW_{ij}}{n} \sum_{k=1, k \neq j}^p W_{ik} v_k\right)\right\} \leq \prod_{i=1}^n \mathbb{E}\left\{\exp\left[\frac{t}{2n} \left\{W_{ij}^2 + \left(\sum_{k=1, k \neq j}^p W_{ik} v_k\right)^2\right\}\right]\right\}.$$

Then, using the Cauchy–Schwarz inequality, we obtain

$$\mathbb{E}\{\exp(t\eta'_j)\} \leq \prod_{i=1}^n \left(\mathbb{E}\left\{\exp\left(\frac{tW_{ij}^2}{n}\right)\right\} \mathbb{E}\left[\exp\left\{\frac{t}{n} \left(\sum_{k=1, k \neq j}^p W_{ik} v_k\right)^2\right\}\right] \right)^{1/2}.$$

Recall that assumption 3 implies that both W_{ij} and $\sum_{k=1, k \neq j}^p W_{ik} v_k$ are σ_* sub-Gaussian. Consequently, in view of expression (24), their squared values are $(\gamma_0(\sigma_*), t_0(\sigma_*))$ subexponential, which yields

$$\mathbb{E}\{\exp(t\eta'_j)\} \leq \prod_{i=1}^n \exp\left\{\frac{\gamma_0(\sigma_*)^2}{2} \left(\frac{t}{n}\right)^2\right\} = \exp\left\{\frac{\gamma_0(\sigma_*)^2 t^2}{2n}\right\}, \quad |t| \leq t_0(\sigma_*)n.$$

Set $\gamma_2 = \gamma_0(\sigma_*)$ and $t_2 = t_0(\sigma_*)$. The last display states that η'_j is $(\gamma_2/n^{1/2}, t_2 n)$ subexponential. This implies the tail bound

$$\mathbb{P}(|\eta'_j| \geq \delta) \leq 2 \max[\exp\{-n\delta^2/(2\gamma_2^2)\}, \exp(-\delta t_2 n/2)],$$

for any $\delta > 0$. This, together with equation (25) and the union bound, yields result (22).

Appendix B: Proofs of theorem 1 and theorem 2

Set for brevity $\delta_i = \delta_i(\varepsilon)$, $\delta'_i = \delta'_i(\varepsilon)$ and $b = b(\varepsilon)$. We first prove some preliminary lemmas.

Lemma 3. Let assumptions 1–4 hold. Then, with probability at least $1 - 6\varepsilon$, the pair $(\theta, t) = (\theta^*, |\theta^*|_2)$ belongs to the feasible set of the minimization problem (9).

Proof. First, note that $Z^T(y - Z\theta^*) + n\hat{D}\theta^*$ is equal to

$$-X^T W \theta^* + X^T \xi + W^T \xi - (W^T W - \text{Diag}\{W^T W\})\theta^* - (\text{Diag}\{W^T W\} - nD)\theta^* + n(\hat{D} - D)\theta^*.$$

By definition of δ_i and b , with probability at least $1 - 4\varepsilon$, we have

$$\left| \frac{1}{n} X^T \xi \right|_\infty + \left| \frac{1}{n} W^T \xi \right|_\infty \leq \delta_2 + \delta_3, \quad (26)$$

$$\left| \left(\frac{1}{n} \text{Diag}\{W^T W\} - D \right) \theta^* \right|_\infty \leq \left| \frac{1}{n} \text{Diag}\{W^T W\} - D \right|_\infty |\theta^*|_\infty \leq \delta_5 |\theta^*|_2, \quad (27)$$

$$|(\hat{D} - D)\theta^*|_\infty \leq b|\theta^*|_\infty \leq b|\theta^*|_2, \quad (28)$$

where in inequalities (27) and (28) we have used that the matrices considered are diagonal. Also, by lemma 2, with probability at least $1 - 2\varepsilon$, we have

$$\left| \frac{1}{n} X^T W \theta^* \right|_{\infty} \leq \delta'_1 |\theta^*|_2, \quad (29)$$

$$\left| \frac{1}{n} (W^T W - \text{Diag}\{W^T W\}) \theta^* \right|_{\infty} \leq \delta'_4 |\theta^*|_2. \quad (30)$$

Combining the decomposition of $Z^T(y - Z\theta^*) + n\hat{D}\theta^*$ together with inequalities (26)–(30), we find that

$$\left| \frac{1}{n} Z^T(y - Z\theta^*) + \hat{D}\theta^* \right|_{\infty} \leq \mu |\theta^*|_2 + \tau,$$

with probability at least $1 - 6\varepsilon$, which implies the lemma. $\square$

We now give two lemmas which will be crucial in the proof of our main theorem on the accuracy of the conic-programming-based estimator (theorem 1).

Lemma 4. Let assumptions 1–4 hold. Let $J = \{j: \theta_j^* \neq 0\}$. Then, with probability at least $1 - 6\varepsilon$ (on the same event as in lemma 3), we have

$$|(\hat{\theta} - \theta^*)_{J^c}|_1 \leq (1 + \lambda) |(\hat{\theta} - \theta^*)_J|_1, \quad (31)$$

$$\hat{t} \leq (1/\lambda) |\hat{\theta} - \theta^*|_1 + |\theta^*|_2. \quad (32)$$

Proof. Set $\Delta = \hat{\theta} - \theta^*$. On the event of lemma 3, $(\theta^*, |\theta^*|_2)$ belongs to the feasible set of the minimization problem (9). Consequently,

$$|\hat{\theta}|_1 + \lambda |\hat{\theta}|_2 \leq |\hat{\theta}|_1 + \lambda \hat{t} \leq |\theta^*|_1 + \lambda |\theta^*|_2. \quad (33)$$

This implies that $|\Delta_{J^c}|_1 \leq |\Delta_J|_1 + \lambda(|\theta^*|_2 - |\hat{\theta}|_2) \leq |\Delta_J|_1 + \lambda|\Delta_J|_2 \leq (1 + \lambda)|\Delta_J|_1$, and inequality (31) follows. To prove inequality (32), it suffices to note that expression (33) implies that

$$\lambda \hat{t} \leq |\theta^*|_1 - |\hat{\theta}|_1 + \lambda |\theta^*|_2 \leq |\hat{\theta} - \theta^*|_1 + \lambda |\theta^*|_2. \quad \square$$

Lemma 5. Let assumptions 1–4 hold. Then, on a subset of the event of lemma 3 having probability at least $1 - 8\varepsilon$, we have

$$\left| \frac{1}{n} X^T X(\hat{\theta} - \theta^*) \right|_{\infty} \leq \mu_1 |\theta^*|_2 + \mu_2 |\hat{\theta} - \theta^*|_1 + \tau_1, \quad (34)$$

where $\mu_1 = \mu + b + \delta'_1 + \delta'_4 + \delta_5$, $\mu_2 = \mu/\lambda + b + 2\delta_1 + \delta_4 + \delta_5$ and $\tau_1 = \tau + \delta_2 + \delta_3$.

Proof. Throughout the proof, we assume that we are on the event of probability at least $1 - 6\varepsilon$ where inequalities (26)–(30) hold and $(\theta^*, |\theta^*|_2)$ belongs to the feasible set of the minimization problem (9). We have that $|(1/n)X^T X\Delta|_{\infty}$ is smaller than

$$\left| \frac{1}{n} Z^T(Z\hat{\theta} - y) - \hat{D}\hat{\theta} \right|_{\infty} + \left| \left(\frac{1}{n} Z^T W - D \right) \hat{\theta} \right|_{\infty} + |(\hat{D} - D)\hat{\theta}|_{\infty} + \left| \frac{1}{n} Z^T \xi \right|_{\infty} + \left| \frac{1}{n} W^T X \Delta \right|_{\infty}.$$

Using the fact that $(\hat{\theta}, \hat{t})$ belongs to the feasible set of the minimization problem (9) together with inequality (32), we obtain

$$\left| \frac{1}{n} Z^T(Z\hat{\theta} - y) - \hat{D}\hat{\theta} \right|_{\infty} \leq \mu \hat{t} + \tau \leq \frac{\mu}{\lambda} |\hat{\theta} - \theta^*|_1 + \mu |\theta^*|_2 + \tau.$$

Therefore, $|(1/n)X^T X\Delta|_{\infty}$ does not exceed

$$\frac{\mu}{\lambda} |\hat{\theta} - \theta^*|_1 + \mu |\theta^*|_2 + \tau_1 + \left| \left(\frac{1}{n} Z^T W - D \right) \hat{\theta} \right|_{\infty} + |(\hat{D} - D)\hat{\theta}|_{\infty} + \left| \frac{1}{n} W^T X \Delta \right|_{\infty}.$$

We now bound the last expression by using the fact that $\hat{\theta} = \theta^* + \Delta$ together with assumption 4 and inequality (28). This gives

$$\begin{aligned} \left| \frac{1}{n} X^T X \Delta \right|_{\infty} &\leq \left(\frac{\mu}{\lambda} + b \right) |\Delta|_1 + (\mu + b) |\theta^*|_2 + \tau_1 + \left| \left(\frac{1}{n} Z^T W - D \right) \theta^* \right|_{\infty} \\ &\quad + \left| \left(\frac{1}{n} Z^T W - D \right) \Delta \right|_{\infty} + \left| \frac{1}{n} W^T X \Delta \right|_{\infty}. \end{aligned} \quad (35)$$

Note that

$$\begin{aligned} \left| \left(\frac{1}{n} Z^T W - D \right) \Delta \right|_{\infty} &\leq \left| \frac{1}{n} Z^T W - D \right|_{\infty} |\Delta|_1 \\ &\leq \left(\left| \frac{1}{n} (W^T W - \text{Diag}\{W^T W\}) \right|_{\infty} + \left| \frac{1}{n} \text{Diag}\{W^T W\} - D \right|_{\infty} + \left| \frac{1}{n} X^T W \right|_{\infty} \right) |\Delta|_1. \end{aligned}$$

Therefore,

$$\left| \left(\frac{1}{n} Z^T W - D \right) \Delta \right|_{\infty} \leq (\delta_1 + \delta_4 + \delta_5) |\Delta|_1, \quad (36)$$

with probability at least $1 - 8\varepsilon$ (since we intersect the initial event of probability at least $1 - 6\varepsilon$ with the event of probability at least $1 - 2\varepsilon$ where the bounds δ_1 and δ_4 hold for the corresponding terms). Next, on the same event of probability at least $1 - 8\varepsilon$,

$$\left| \frac{1}{n} W^T X \Delta \right|_{\infty} \leq \left| \frac{1}{n} X^T W \right|_{\infty} |\Delta|_1 \leq \delta_1 |\Delta|_1. \quad (37)$$

Finally, in view of lemma 2 and inequality (27), on the initial event of probability at least $1 - 6\varepsilon$,

$$\begin{aligned} \left| \left(\frac{1}{n} Z^T W - D \right) \theta^* \right|_{\infty} &\leq \left| \frac{1}{n} (W^T W - \text{Diag}\{W^T W\}) \theta^* \right|_{\infty} + \left| \left(\frac{1}{n} \text{Diag}\{W^T W\} - D \right) \theta^* \right|_{\infty} + \left| \frac{1}{n} X^T W \theta^* \right|_{\infty} \\ &\leq (\delta'_1 + \delta'_4 + \delta_5) |\theta^*|_2. \end{aligned} \quad (38)$$

To complete the proof, it suffices to plug inequalities (36)–(38) in expression (35) and to set $\mu_1 = \mu + b + \delta'_1 + \delta'_4 + \delta_5$ and $\mu_2 = \mu/\lambda + b + 2\delta_1 + \delta_4 + \delta_5$. $\square$

We are ready to give the proof of theorem 1. Throughout the proof, we assume that we are on the event of probability at least $1 - 8\varepsilon$ of lemma 5 where the results of lemmas 3, 4 and 5 hold. Property (31) in lemma 4 implies that Δ is in the cone $C_J(1 + \lambda)$. Therefore, by definition of l_q -sensitivity and lemma 5, we have

$$\kappa_q(s, 1 + \lambda) |\Delta|_q \leq \left| \frac{1}{n} X^T X \Delta \right|_{\infty} \leq \mu_1 |\theta^*|_2 + \mu_2 |\Delta|_1 + \tau_1.$$

Furthermore, using again inequality (31), we have

$$\begin{aligned} |\Delta|_1 &= |\Delta_{J^c}|_1 + |\Delta_J|_1 \leq (2 + \lambda) |\Delta_J|_1 \\ &\leq (2 + \lambda) s^{1-1/q} |\Delta_J|_q \leq (2 + \lambda) s^{1-1/q} |\Delta|_q. \end{aligned}$$

It follows that

$$\{\kappa_q(s, 1 + \lambda) - (2 + \lambda) \mu_2 s^{1-1/q}\} |\Delta|_q \leq \mu_1 |\theta^*|_2 + \tau_1.$$

Further, the assumption $s \leq c_1 (\lambda^{-1} + \lambda)^{-1} \{n/\log(p/\varepsilon)\}^{1/2}$ implies that

$$[c - \mu_2 c_1 (2 + \lambda) (\lambda^{-1} + \lambda)^{-1} \{n/\log(p/\varepsilon)\}^{1/2}] s^{-1/q} |\Delta|_q \leq \mu_1 |\theta^*|_2 + \tau_1.$$

Recall that $\mu_2 \leq (1 + \lambda^{-1}) a \{\log(p/\varepsilon)/n\}^{1/2}$, where $a > 0$ is a constant. Therefore, inequality (13) follows if $(2 + \lambda)(1 + \lambda^{-1})(\lambda^{-1} + \lambda)^{-1} c_1 a < c/2$. Since $(2 + \lambda)(1 + \lambda^{-1})(\lambda^{-1} + \lambda)^{-1} \leq 5$ we have that $c_1 < c/(10a)$ yields inequality (13). To prove inequality (14), write first

$$\frac{1}{n} |X\Delta|_2^2 \leq \frac{1}{n} |X^T X \Delta|_\infty |\Delta|_1.$$

Next note that, from inequality (13), we have

$$|\Delta|_1 \leq C_s \left\{ \frac{\log(c' p/\varepsilon)}{n} \right\}^{1/2} (|\theta^*|_2 + 1) \quad (39)$$

and recall that, from lemma 5,

$$\left| \frac{1}{n} X^T X \Delta \right|_\infty \leq C \left\{ \frac{\log(c' p/\varepsilon)}{n} \right\}^{1/2} (1 + |\theta^*|_2 + |\Delta|_1). \quad (40)$$

Furthermore, from inequalities (39) and (12) we also have that $|\Delta|_1 \leq C(|\theta^*|_2 + 1)$, for some constant $C > 0$. Then inequality (14) is easily deduced. This ends the proof of theorem 1.

We now give the proof of theorem 2. We place ourselves in the same framework as in the proof of theorem 1. By the definition of the estimator, $|\Delta|_1 \leq |\hat{\theta}|_1 + |\theta^*|_1 \leq (|\theta^*|_1 + \lambda|\theta^*|_2) + |\theta^*|_1 \leq (2 + \lambda)|\theta^*|_1$, where we have used that $|\theta^*|_2 \leq |\theta^*|_1$. This and lemma 5 yield

$$\left| \frac{1}{n} X^T X \Delta \right|_\infty \leq \mu_1 |\theta^*|_2 + \mu_2 |\Delta|_1 + \tau_1 \leq \{\mu_1 + (2 + \lambda)\mu_2\} |\theta^*|_1 + \tau_1.$$

Therefore, arguing as in the proof of theorem 1, we find that

$$\kappa_q(s, 1 + \lambda) |\Delta|_q \leq \{\mu_1 + (2 + \lambda)\mu_2\} |\theta^*|_1 + \tau_1,$$

which implies inequality (16) since $\mu_2 \leq (1 + \lambda^{-1})a\{\log(p/\varepsilon)/n\}^{1/2}$ for some constant $a > 0$. To prove inequality (17), we note that, by definition of κ_{pr} and the fact that $\Delta \in C_J(1 + \lambda)$,

$$\frac{\kappa_{\text{pr}}^2(s, 1 + \lambda)}{n} |X\Delta|_2^2 \leq \frac{1}{n} |X^T X \Delta|_\infty^2 \leq [\{\mu_1 + (2 + \lambda)\mu_2\} |\theta^*|_1 + \tau_1]^2.$$

Finally, since $|\Delta|_1 \leq (2 + \lambda)|\theta^*|_1$ and $\mu_2 \leq (1 + \lambda^{-1})a\{\log(p/\varepsilon)/n\}^{1/2}$, inequality (18) follows from

$$\frac{1}{n} |X\Delta|_2^2 \leq \frac{1}{n} |X^T X \Delta|_\infty |\Delta|_1 \leq [\{\mu_1 + (2 + \lambda)\mu_2\} |\theta^*|_1 + \tau_1] (2 + \lambda) |\theta^*|_1.$$

This concludes the proof of theorem 2.

Appendix C: Proof of theorem 3

Again, throughout, we denote by c a positive constant which may vary from line to line. To derive the lower bounds, we apply theorem 2.7 in Tsybakov (2009). Thus, we define a finite set of ‘hypotheses’ included in $B_0(s) \cap S_2(R)$. For this, we first introduce

$$\mathcal{M} = \{x \in \{0, 1\}^{p-1} : \rho_H(\mathbf{0}, x) = s - 1\},$$

where ρ_H denotes the Hamming distance between elements of $\{0, 1\}^{p-1}$, and $\mathbf{0}$ is the zero vector. Then, there is a subset $\mathcal{M}'$ of $\mathcal{M}$ such that, for any x and x' in $\mathcal{M}'$ with $x \neq x'$, we have $\rho_H(x, x') > s/16$ and, moreover,

$$\log |\mathcal{M}'| \geq c'_1 s \log \left(\frac{p}{s} \right)$$

for some absolute constant $c'_1 > 0$. Indeed, this follows from the Varshamov–Gilbert bound (see lemma 2.9 in Tsybakov (2009)) if $s - 1 > (p - 1)/2$ and from lemma A.3 in Rigollet and Tsybakov (2011) if $s - 1 \leq (p - 1)/2$.

We denote by ω'_j the elements of the finite set $\mathcal{M}'$. For $j = 1, \dots, |\mathcal{M}'|$, we define vectors $\omega_j \in \{0, 1\}^p$ with components $\omega_{j1} = 0$ and $\omega_{jk} = \omega'_{j(k-1)}$ for $k \geq 2$, where ω_{jk} is the k th component of ω_j . We also define ω_0 as the vector in $\{0, 1\}^p$ with all components equal to 0 except the first, which is equal to 1.

We now define the set of ‘hypotheses’ $(\bar{\omega}_j, j = 0, \dots, |\mathcal{M}'| + 1)$, where $\bar{\omega}_0 = R\omega_0$, and

$$\bar{\omega}_j = \frac{R}{\{1 + \gamma^2(s - 1)\}^{1/2}} (\omega_0 + \gamma\omega_j), \quad j = 1, \dots, |\mathcal{M}'| + 1.$$

Here, γ is a positive parameter to be defined. Note that the sparsity of $\bar{\omega}_j$ is equal to s and that $\|\bar{\omega}_j\|_2 = R$. Thus all $\bar{\omega}_j$ belong to $B_0(s) \cap S_2(R)$. Let

$$\tilde{\Sigma} = \sigma_*^2 (\Sigma + \sigma_*^2 I_{p \times p})^{-1}$$

and

$$\Gamma = I_{p \times p} - \tilde{\Sigma}.$$

For $\theta \in \mathbb{R}^p$, we set $c_\theta = \theta^T \Gamma \theta$ and we write $\mathcal{K}(\mathbb{P}, \mathbb{Q})$ for the Kullback–Leibler divergence between two probability measures $\mathbb{P}$ and $\mathbb{Q}$. For $j \geq 1$, by lemma 9 in the on-line supplementary material, we have

$$\begin{aligned} \mathcal{K}(\mathbb{P}_{\bar{\omega}_j}, \mathbb{P}_{\bar{\omega}_0}) &\leq \frac{cn}{1+R^2} \left(R^2 \left[\frac{\{1+\gamma^2(s-1)\}^{1/2}-1}{\{1+\gamma^2(s-1)\}^{1/2}} \right]^2 + \frac{R^2}{1+\gamma^2(s-1)} \gamma^2 s + |c_{\bar{\omega}_j} - c_{\bar{\omega}_0}| \right) \\ &\leq \frac{cn}{1+R^2} \left\{ \frac{\gamma^2 R^2 s}{1+\gamma^2(s-1)} + |c_{\bar{\omega}_j} - c_{\bar{\omega}_0}| \right\}. \end{aligned}$$

Now,

$$\begin{aligned} |c_{\bar{\omega}_j} - c_{\bar{\omega}_0}| &= |\bar{\omega}_j^T \Gamma \bar{\omega}_j - \bar{\omega}_0^T \Gamma \bar{\omega}_0| = \left| R^2 \bar{\omega}_0^T \Gamma \bar{\omega}_0 - \frac{R^2}{1+\gamma^2(s-1)} \bar{\omega}_0^T \Gamma \bar{\omega}_0 - \frac{\gamma^2 R^2}{1+\gamma^2(s-1)} \bar{\omega}_j^T \Gamma \bar{\omega}_j \right| \\ &\leq \frac{\gamma^2 R^2}{1+\gamma^2(s-1)} \{ (s-1) \bar{\omega}_0^T \Gamma \bar{\omega}_0 + \bar{\omega}_j^T \Gamma \bar{\omega}_j \} \leq \frac{2\lambda_{\max} \gamma^2 R^2 (s-1)}{1+\gamma^2(s-1)}, \end{aligned}$$

where $\lambda_{\max}$ denotes the largest eigenvalue of Γ and the last inequality is due to the fact that $\|\bar{\omega}_0\|_2 = 1$ and $\|\bar{\omega}_j\|_2^2 = s-1$. Combining the last two displays yields

$$\mathcal{K}(\mathbb{P}_{\bar{\omega}_j}, \mathbb{P}_{\bar{\omega}_0}) \leq c'_2 n \gamma^2 s \frac{R^2}{1+R^2},$$

where $c'_2 > 0$ is a constant that does not depend on s , p , R and n . Now, taking

$$\gamma = \left\{ \frac{c'_1}{16c'_2 n} \frac{1+R^2}{R^2} \log\left(\frac{p}{s}\right) \right\}^{1/2}, \quad (41)$$

we obtain, for all j ,

$$\mathcal{K}(\mathbb{P}_{\bar{\omega}_j}, \mathbb{P}_{\bar{\omega}_0}) \leq \frac{1}{16} \log |\mathcal{M}'|.$$

Next, for j and j' both different from 0,

$$\|\bar{\omega}_j - \bar{\omega}_{j'}\|_q = \frac{R\gamma}{\{1+\gamma^2(s-1)\}^{1/2}} \left(\sum_{k=1}^{p-1} |\omega_{jk} - \omega_{j'k}|^q \right)^{1/q} \geq cs^{1/q} \frac{R\gamma}{\{1+\gamma^2(s-1)\}^{1/2}}$$

and, for $j \neq 0$,

$$\|\bar{\omega}_j - \bar{\omega}_0\|_q \geq \frac{R\gamma \|\bar{\omega}_j\|_q}{\{1+\gamma^2(s-1)\}^{1/2}} \geq cs^{1/q} \frac{R\gamma}{\{1+\gamma^2(s-1)\}^{1/2}}.$$

The definition of γ in equation (41) and the conditions in theorem 3 imply that, for any j and j' ,

$$\|\bar{\omega}_j - \bar{\omega}_{j'}\|_q \geq cs^{1/q} R\gamma \geq cs^{1/q} (R+1) \left\{ \frac{\log(p/s)}{n} \right\}^{1/2}.$$

We can now apply theorem 2.7 in Tsybakov (2009) to obtain the result.

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Supporting information

Additional ‘supporting information’ may be found in the on-line version of this article:
‘Supplementary material’.